

Data Cleaning

SCIKIT LEARN to impute missing value

Download the package

```
pip install -U scikit-learn
```

Now import library file.

In scikit learn there is library called simpleimputer for impute missing value

SimpleImputer

Replace missing values using a descriptive statistic (e.g. mean, median, or most frequent) along each column, or using a constant value.

Methods

`fit(X[, y])` Fit the imputer on x.

`fit_transform(X[, y])` Fit to data, then transform it.

`fit(X, y=None)[source]`

Fit the imputer on x.

Parameters:

X{array-like, sparse matrix}, shape (n_samples, n_features)

Input data, where `n_samples` is the number of samples and `n_features` is the number of features.

yIgnored

Not used, present here for API consistency by convention.

Returns:

selfobject

Fitted estimator.

`fit_transform(X, y=None, **fit_params)[source]`

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Fit to data, then transform it.

Fits transformer to `X` and `y` with optional parameters `fit_params` and returns a transformed version of `X`.

Parameters:

Xarray-like of shape (n_samples, n_features)

Input samples.

yarray-like of shape (n_samples,) or (n_samples, n_outputs), default=None

Target values (None for unsupervised transformations).

*****fit_paramsdict***

Additional fit parameters.

Returns:

X_newndarray array of shape (n_samples, n_features_new)

Transformed array.

Now import the Library

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer
```

now we have two dataset read the train & test dataset

```
train=pd.read_csv('train.csv')
test=pd.read_csv('test.csv')
print('train shape',train.shape)
print('test shape',test.shape)
```

now see that train dataset has 81 columns & test dataset has 80 column

```
train.head()
```

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which means in train dataset has salesprice columns is predicted that's why is missing in test dataset

now we need x_train & y_train for predicted

```
X_train=train.drop(columns='SalePrice')
y_train=train['SalePrice']
print('train shape',X_train.shape)
print('test shape',y_train.shape)
```

we have store salesprice variable in x_train to predict y feature

Numerical Missing Value Imputation

Now find only numerical columns

```
num_col=X_train.select_dtypes(include=['int64','float64']).columns
num_col
```

now find any missing value in this columns

```
X_train[num_col].isnull().sum()
```

Now we use mean strategy from sklearn to impute values using SimpleImputer class

We have by default class value impute is mean ..you can choose any

```
impute_mean=SimpleImputer(strategy='mean')
```

we create mean of data. If you want to use this method need dataset.

Now we need to use fit() to use statistical dataset

We need only mean for numerical dataset

Fit() check mean value of dataset and store the mean value

```
impute_mean.fit(X_train[num_col])
```

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now we check how the value is going on

```
impute_mean.statistics_
```

executing this we saw all the mean value for numeric columns but it is not applied on dataset.

For filling mean value on missing data we need to use Transform() method for particular dataset.

```
impute_mean.transform(X_train[num_col])
```

transform method is given 2Darray ..

we fill this value on to our dataset

```
X_train[num_col]=impute_mean.transform(X_train[num_col])  
test[num_col]=impute_mean.transform(test[num_col])
```

Note :- keep in mind that when we fill missing value in x_train dataset at that time also fill same mean on test dataset.

We don't create mean of test dataset.

Now find is there any null values present in this dataset.

```
X_train[num_col].isnull().sum()
```

Also check on test dataset

```
test[num_col].isnull().sum()
```

for arbitrary value

we use different strategy for arbitrary value.

Use constat strategy for your value.

```
impute_mean=SimpleImputer(strategy='constant',fill_value=90)
```

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now find categorical missing value imputation

find categorical data columns

```
cat_col=X_train.select_dtypes(include="O").columns  
num_col
```

now find isthere any categorical data missing

```
X_train[cat_col].isnull().sum()
```

Now impute mode strategy to fill value inplace of missing value

```
imputer_mode=SimpleImputer(strategy='most_frequent')  
imputer_mode
```

now apply imputermode value using fit method to fill value

```
imputer_mode.fit(X_train[cat_col])
```

now find the mode value of perticular each dataset

```
imputer_mode.statistics_
```

now need transform method to fill value using mode.

```
X_train[cat_col]=imputer_mode.transform(X_train[cat_col])  
test[cat_col]=imputer_mode.transform(test[cat_col])
```

now check is there any missing value present in the dataset

```
X_train[cat_col].isnull().sum()
```

Also check on test dataset

```
test[cat_col].isnull().sum()
```

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now find is there any missing value on whole dataset

```
X_train.isnull().sum().sum()
```

Now clearly see that there isn't any missing value present in the dataset.